ALICE L WALTON SCHOOL OF MEDICINE

Nonopioid Analgesics – Acetaminophen

Last updated May 27, 2026 11 min read ScholarRx
Adapted by: Diptiman Bose, Trager Hintze
Editors: Jordan Finn, Stephen Nix, Russell Johnson
Adapted from: Nonopioid Analgesics

Listen to this Brick
0:00 / 20:21 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Compare the pharmacological characteristics and side effect profiles of acetaminophen and NSAIDs, emphasizing their clinical implications.
2. Discuss the metabolic pathway of acetaminophen and explain the factors which influence its metabolism.
3. Identify the risk factors associated with acetaminophen-induced hepatotoxicity and explain the pathophysiologic mechanisms leading to liver toxicity.
4. Explain the mechanism by which N-acetylcysteine (NAC) functions as an antidote in the context of acetaminophen overdose.

CASE CONNECTION

AL is a 23-year-old male college student who has had a cold and 'felt like he had a fever' for the last 48 hours. He did not measure his temperature. He knows that several over-the-counter cold and flu medications are available for symptom relief. He had some acetaminophen (500 mg) in his medicine cabinet. He took two of them. He also went to the local pharmacy and purchased a combination product of acetaminophen (650 mg/ 30 mL)/ doxylamine (12.5 mg/ 30 mL)/ dextromethorphan (20 mg/ 30 mL) (300 mL). By the end of the day, he drank all 300 ml and had taken 2 more acetaminophen tablets (500 mg). He started feeling better and decided to go out drinking with his classmates.

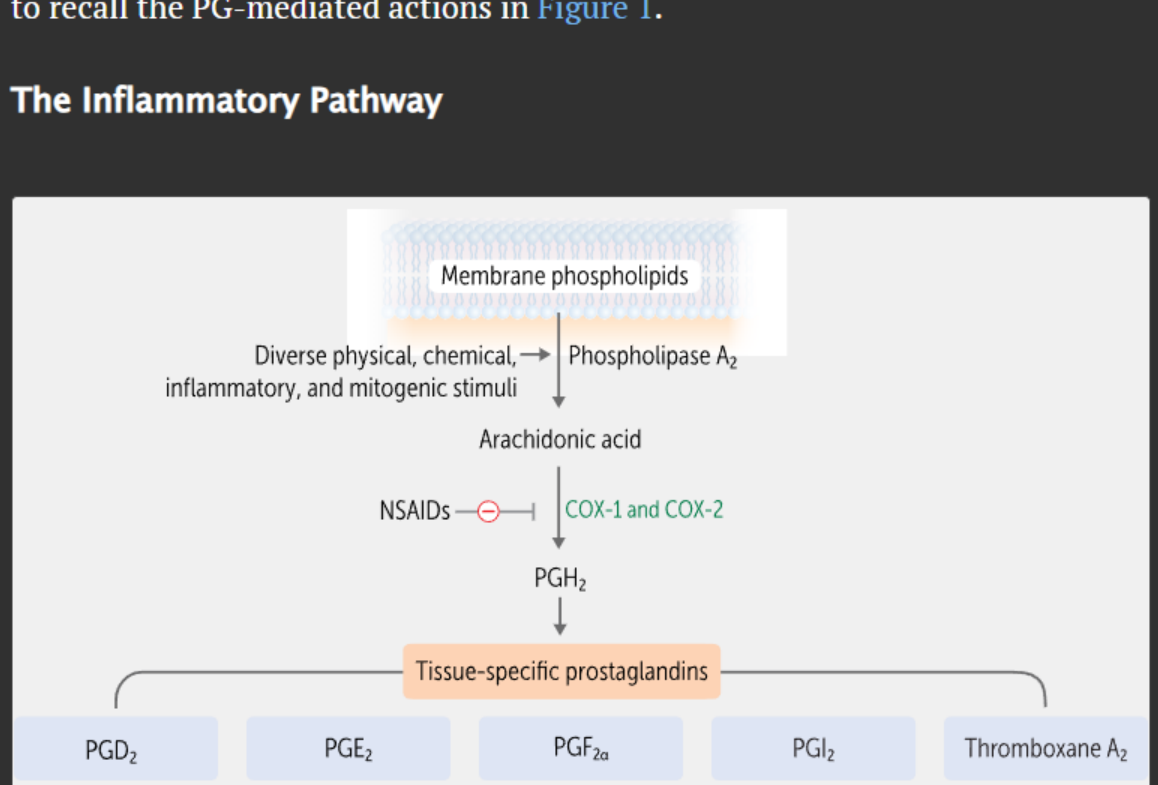
Is AL at risk of any serious adverse effects from taking these medications at once? Are there any risks with drinking alcohol while taking these OTC products?

GO TO CONCLUSION ↓

Nonsteroidal Anti–inflammatory Drugs

We have discussed th[illegible] NSAID and Aspirin bricks. Before we proceed to discuss the mechanism of acetaminophen, it is important to recall the PG-mediated actions in Figure 1.

The Inflammatory Pathway


Membrane phospholipids
Diverse physical, chemical, inflammatory, and mitogenic stimuli
Phospholipase A_2
Arachidonic acid
NSAIDs
COX-1 and COX-2
PGH_2
Tissue-specific prostaglandins
PGD_2
Mast cells, brain, airways
PGE_2
Pain receptors, brain, kidneys, blood vessels, platelets
$PGF_{2\alpha}$
Pain receptors, uterus, airways, eyes
PGI_2
Endothelium, blood vessels, platelets, kidney, brain
Thromboxane A_2
Platelets, blood vessels, macrophages, kidney

QUIZ

Figure 1

Credit: @ScholarRx

COX-1 and COX-2 convert which fatty acid into prostaglandin intermediates?

What is the difference between COX-1 and COX-2?

What is the mechanism of action of acetaminophen? (LO#1)

Like NSAIDs, acetam[illegible] used nonopioid analgesic. Acetaminophen is a **weak non-competitive reversible inhibitor** of COX-1 and -2 function in the CNS and has weak peripheral effects on COX function. Although acetaminophen can inhibit COX activity, its effects are primarily central, which helps explain its analgesic and antipyretic effects and its weak peripheral anti-inflammatory activity. However, in the CNS, the acetaminophen is oxidized by the brain CYP 450 system and the oxidized form of the drug inhibits COX activity. The enzyme function is inhibited at the peroxidase catalytic site rather than the COX catalytic site.

Its limited peripheral COX inhibition helps explain why acetaminophen has analgesic and antipyretic effects but weak anti-inflammatory effects. Acetaminophen can also increase the pain threshold against a pain stimulus, thus exhibiting **analgesic effects**. This mechanism also distinguishes acetaminophen from the NSAIDs, hence *not classified as an NSAID*. Higher levels of peroxidase in platelets block acetaminophen from acting, thus it does not act as an antithrombotic agent.

Why does acetaminophen have weak anti-inflammatory effects?

What are some adverse effects of acetaminophen? (LO#1)

Acetaminophen is generally well tolerated when taken at recommended doses and should not exceed 4 g/day in most adults. Acetaminophen causes fewer adverse effects than NSAIDs, usually mild nausea, GI upset, rash and other hypersensitivity reactions, itching, and headache. Rare but serious adverse effects include rash/ hypersensitivity reaction, abnormal bruising, jaundice, and acute renal failure. Hepatotoxicity remains a major concern with acetaminophen. Intentional overdose associated with suicidal ideation is common, and unintentional overdoses are also common. Unintentional overdoses can occur because acetaminophen is widely available OTC and is included in many single-ingredient and combination products. Additionally, acetaminophen can be found in combination products with opioids, and overdose from these combination products is also significant. Because acetaminophen is metabolized by the liver, caution should be used in patients with chronic alcohol use or hepatitis. Mortality rates are lower with early detection; however, if acute liver failure occurs, the mortality rates increase to 28%.

What is the maximum recommended daily dose of acetaminophen?

How is acetaminophen metabolized? (LO#2)

Because the drug is so widely used and available OTC, it is commonly involved in intentional or unintentional overdose. In an **acetaminophen overdose**, patients present with abdominal pain, intractable nausea and vomiting, and a tender hepatic edge on physical exam within the first 24-72 hours. Serum transaminases can rise rapidly, and severe cases can progress to acute liver failure.

Acetaminophen metabolism (Figure 2) occurs mainly in the liver:

- Acetaminophen has good bioavailability and peak concentrations can be reached within 30-60 min. The half-life is approximately 2 hours, and acetaminophen distributes widely into body fluids.
- Suggested revision: "Most acetaminophen undergoes phase II conjugation: approximately 45%–55% via glucuronidation and 30%–40% via sulfation.
- Another route, mediated by the intracellular cytochrome P450 enzyme system (pathway on the left), leads to the formation of the potentially toxic **N-acetyl-p-benzoquinone imine (NAPQI).** This is usually only 10% of acetaminophen excretion, and the potentially toxic NAPQI is normally conjugated with g[illegible]

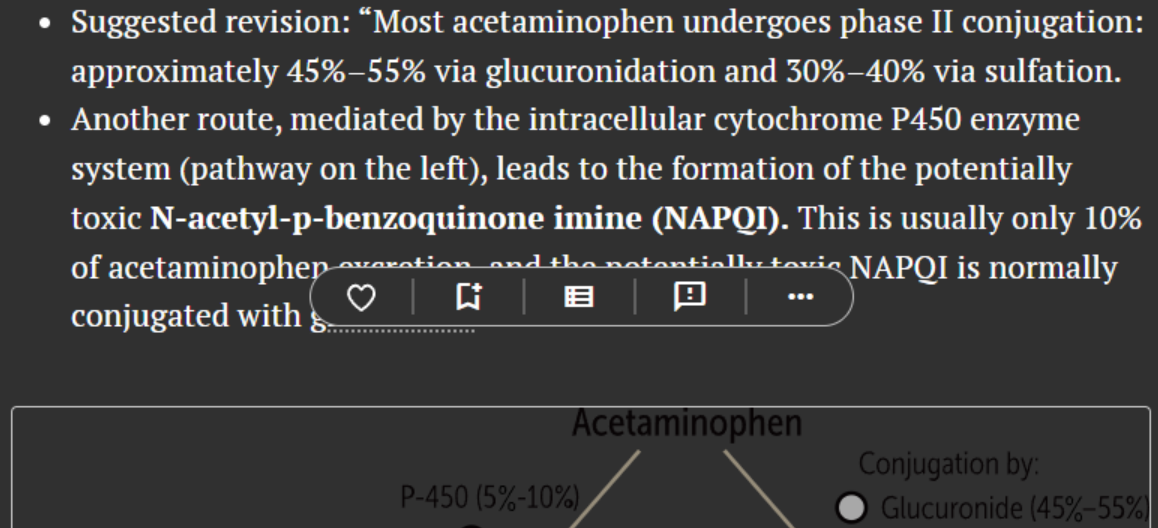


Figure 2

When toxic amounts of acetaminophen are ingested (in an overdose), the sulfation and glucuronidation pathways are saturated. The acetaminophen metabolism shifts towards the CYP P450 (CYP2E1) and more NAPQI is formed. NAPQI depletes glutathione and starts to accumulate. The toxic NAPQI covalently bonds with hepatic proteins, forming adducts resulting in oxidative stress and mitochondrial damage to the liver cells (hepatocytes). These products can lead to oxidative hepatic injury and hepatocellular centrilobular necrosis causing acetaminophen-induced hepatotoxicity.

What is the toxic metabolite formed during acetaminophen overdose?

What are the factors that impact acetaminophen metabolism? (LO#3)

Chronic alcohol use induces CYP2E1, increasing conversion of acetaminophen to NAPQI. Chronic alcohol use is also often associated with poor nutrition, which can reduce glutathione reserves. Together, these factors increase hepatotoxicity risk. In a malnourished state, glucuronidation of acetaminophen is reduced, which leads to enhanced microsomal metabolism and increased production of NAPQI, predisposing these patients to liver toxicity. Adults over 40 years of age have a greater risk of acute liver failure, liver transplantation, and death following overdose.

What role [illegible]n-induced hepatotoxicity?

Concomitant use of other medications can also increase the risk of hepatic injury and other adverse effects. Although acetaminophen overall has minimal drug-drug interactions, there are a few interactions to be aware of. Isoniazid (a medication for tuberculosis), carbamazepine (an antiepileptic), and alcohol all increase acetaminophen's risk of hepatotoxicity by increasing the amount of NAPQI produced in the liver. Acetaminophen can also increase the effects of the anticoagulation drug warfarin, potentially leading to increased anticoagulation and risk of bleeding. Thus, a patient's warfarin dose may need to be adjusted.

Which drugs increase acetaminophen's risk of hepatotoxicity?

CLINICAL CORRELATION

You may have heard not to use acetaminophen along with alcohol. This is because one specific P450 enzyme called CYP2E1 is induced by ethanol, leading to more NAPQI formation.

How do you reverse acetaminophen toxicity? (LO#4)

Once the diagnosis of acetaminophen overdose is established (history, serum drug levels), the antidote N-acetylcysteine (NAC) should promptly be given, preferably within the first 16 hours after the overdose. NAC (see Figure 2, left) provides the cysteine, which helps restore glutathione levels, which results in a decrease in toxic NAPQI levels. It has been shown to scavenge free radicals and increase mitochondrial ATP production.

Although not common, the use of NAC could result in adverse reactions. Common adverse reactions include facial flushing (≤8%), urticaria (≤8%), and rash (4%).

What is the primary antidote for acetaminophen overdose?

CLINICAL CORRELATION

The original Rumack-Matthew nomogram relates serum acetaminophen concentration to the time of ingestion as a predictor of hepatotoxicity (Figure 3). Patients with serum acetaminophen concentrations below the "probable hepatic toxicity" line did not develop severe hepatotoxicity. Without antidotal therapy, patients with serum acetaminophen concentrations above the "probable hepatic toxicity" line had a 60 percent incidence of severe hepatotoxicity (AST greater than 1000 IU/L) and a mortality rate of 5 percent. If left untreated with high serum acetaminophen concentrations above the line, joining 300 mcg/mL (1980 micromol/L) at 4 hours and 37.5 mcg/mL (250 micromol/L) at 16 hours ("high hepatic toxicity" or "high-risk ingestion") have a 90 percent incidence of severe hepatotoxicity and a mortality rate of up to 24 percent. High-risk patients may require an increased dose of acetylcysteine and consultation with a regional poison control center or clinical pharmacist.

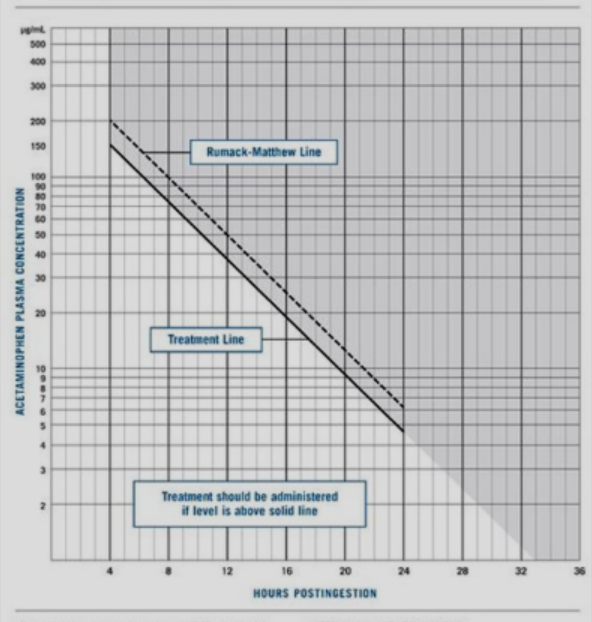


Figure 3. Rumack-Matthew nomogram for acetaminophen overdose

What is the primary function of NAC in acetaminophen overdose?

CASE CONNECTION BACK TO INTRODUCTION ↑

Is AL at risk of any serious adverse effects from taking these medications at once? Are there any risks with drinking alcohol while taking these OTC products?

Acetaminophen can provide symptomatic relief for AL's fever and pain. Acetaminophen provides analgesic and antipyretic effects through central COX inhibition and by increasing the pain threshold. In general, acetaminophen and its combination products that are OTC or prescribed are considered safe. However, as stated in the Brick unintentional overdose from using acetaminophen products are common. Hepatotoxicity associated with acetaminophen use is a major concern. Hence, max/day (4g/day) limits for acetaminophen are provided on the label. AL is under the daily dose limit, but concerns about overdosing on acetaminophen are valid here. We know that acetaminophen can be metabolized to NAPQI and excreted renally. However, excess acetaminophen can increase NAPQI formation, deplete glutathione, and promote hepatic protein adduct formation, leading to hepatocellular injury. Consumption of alcohol increases the susceptibility to hepatic injury with acetaminophen

Summary

- Acetaminophen, unlike NSAIDs, has weaker anti-inflammatory effects and is primarily used for analgesic and antipyretic action
- The mechanism of action of acetaminophen involves weak inhibition of COX-1 and -2 fun[illegible] effects on COX function.
- Adverse effects of acetaminophen are usually mild but can include serious conditions like hepatic damage, especially in cases of overdose.
- Acetaminophen overdose can occur with intentional or unintentional use of the drug
- Acetaminophen-induced hepatotoxicity is caused by the metabolites that deplete glutathione and the production of adducts that can cause oxidative hepatic injury.
- Acetaminophen-induced hepatotoxicity can be reversed by the use of NAC
- Acetaminophen's metabolism mainly occurs in the liver, and in toxic amounts, it can lead to hepatotoxicity.
- While acetaminophen has minimal drug-drug interactions, it can increase the risk of hepatotoxicity when combined with certain medications like Isoniazid, Carbamazepine, and alcohol.

Review Questions

1. Acetaminophen's mechanism of action differs from NSAIDs because it

A. ○ Binds to COX in the peripheral tissues
B. ● Binds at a different COX binding site
C. ○ Binds only to COX-2 in the CNS
D. ○ Binds only to COX-1 in the periphery

Hide Explanation

The correct answer is binds to a different COX binding site (B). Acetaminophen binds to the peroxide site of the enzyme in the CNS. Binding to the COX site in peripheral tissues in similar to NSAID, but is weak (A). Acetaminophen is a non-selective COX inhibitor, so acetaminophen is not a selective COX-1 inhibitor (D) or a COX-2 inhibitor (C).

2. What is a major adverse effect of acetaminophen use?

A. ○ GI upset
B. ○ Rash
C. ○ Nausea
D. ● Hepatotoxicity
E. ○ Headache

Hide Explanation

The correct answer is hepatotoxicity (D). Hepatotoxicity is a major concern due to the potential for liver damage, especially in cases of overdose. Nausea (C) and rash (B) are common but mild adverse effects. GI upset (A) is also a common but less severe issue. Headache (E) is another mild adverse effect.

3. Which medication is most likely to increase acetaminophen's risk of hepatotoxicity?

A. ● Isoniazid
B. ○ Carbamazepine
C. ○ Alcohol
D. ○ Warfarin
E. ○ N-acetylcysteine

Hide Explanation

The correct answer is Isoniazid (A). Isoniazid increases the risk of hepatotoxicity by increasing the amount of NAPQI produced in the liver. Carbamazepine (B) and alcohol (C) also increase NAPQI production, but they are not the most common causes. Warfarin (D) does not increase hepatotoxicity; it increases the risk of bleeding. N-acetylcysteine (E) is an antidote, not a risk factor.

4. What is the toxic metabolite of acetaminophen?

A. ○ Glucuronides
B. ○ Sulfates
C. ● NAPQI
D. ○ Glutathione
E. ○ Transaminases

Hide Explanation

The correct answer is NAPQI (C). NAPQI is the toxic metabolite formed via the cytochrome P450 pathway. Glucuronides (A) and sulfates (B) are non-toxic conjugates. Glutathione (D) is a molecule that detoxifies NAPQI. Transaminases (E) are liver enzymes, not metabolites.

5. Which patient has the least direct risk factor for acetaminophen-induced hepatotoxicity?

A. ○ Acute hepatitis
B. ● Acute interstitial nephritis
C. ○ Chronic alcoholism
D. ○ Cirrhosis

Hide Explanation

The correct answer is acute interstitial nephritis (B). Acetaminophen is contraindicated in severe hepatic dysfunction, as in the case of acute hepatitis (A) and cirrhosis (D). Patients who have chronic alcoholism (C) have a high risk of alcohol-related hepatitis and dysfunction, so they should not receive acetaminophen.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare the pharmacological characteristics and side effect profiles of acetaminophen and NSAIDs, emphasizing their clinical implications.~~
02 ☑ ~~Discuss the metabolic pathway of acetaminophen and explain the factors which influence its metabolism.~~
03 ☑ ~~Identify the risk factors associated with acetaminophen-induced hepatotoxicity and explain the pathophysiologic mechanisms leading to liver toxicity.~~
04 ☐ Explain the mechanism by which N-acetylcysteine (NAC) functions as an antidote in the context of acetaminophen overdose.

acetaminophen.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare the pharmacological characteristics and side effect profiles of acetaminophen and NSAIDs, emphasizing their clinical implications.~~
02 ☑ ~~Discuss the metabolic pathway of acetaminophen and explain the factors which influence its metabolism.~~
03 ☑ ~~Identify the risk factors associated with acetaminophen-induced hepatotoxicity and explain the pathophysiologic mechanisms leading to liver toxicity.~~
04 ☐ Explain the mechanism by which N-acetylcysteine (NAC) functions as an antidote in the context of acetaminophen overdose.

Objective Confidence: 3/4